Kickstarter Campaign Analysis

A Data Essay by Bryan Beus

Problem Identification

- Small-business owners sometimes have valuable ideas that require funding in order to become a reality
- Crowdsourcing is when an entrepreneur turns to the open internet for support for their ideas

Problem Identification

- A successful crowdsourcing campaign can help the entrepreneur, but a failed campaign can lead to a loss of time, resources, and social investment

Project Context

- This data-driven essay assists crowdsourcing entrepreneurs through:
  - analytical techniques
  - an open-source dataset
  - predictive modeling

Solution Scope

- The scope of this essay includes tools such as:
  - Web Robots archive
  - Python
  - Jupyter
  - Numpy
  - Matplotlib
  - Scikit-Learn
  - Seaborn

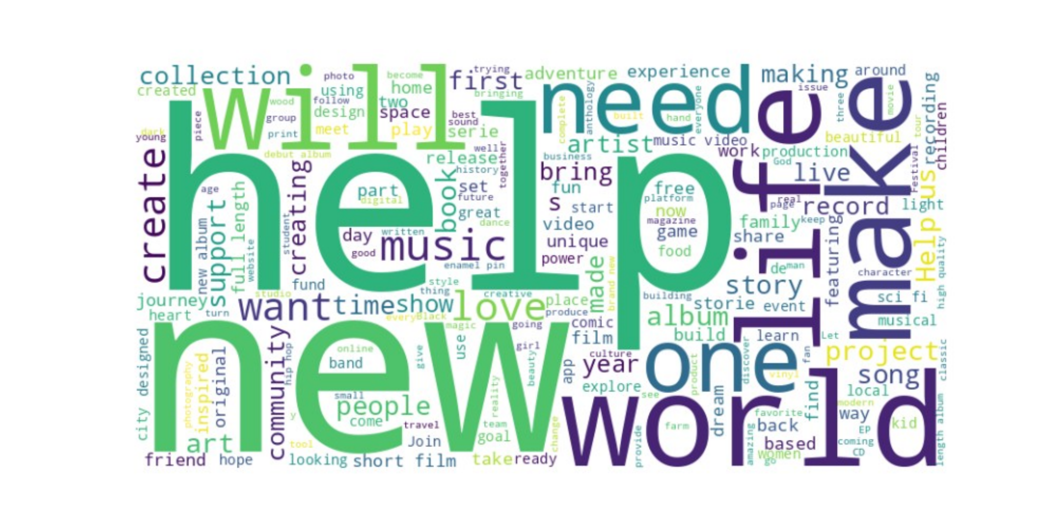
Constraints

- Based on publicly available datasets:
  - Incomplete web-scrapings
  - Selection bias
- Only general categorization
  - Small-business owners may find that their project niche is difficult to relate

Data Wrangling

- Kickstarter dataset
  - Import to Jupyter
  - Convert json to pandas dataframe
  - Remove duplicates, null values
  - Drop unnecessary columns
  - Export as parquet file

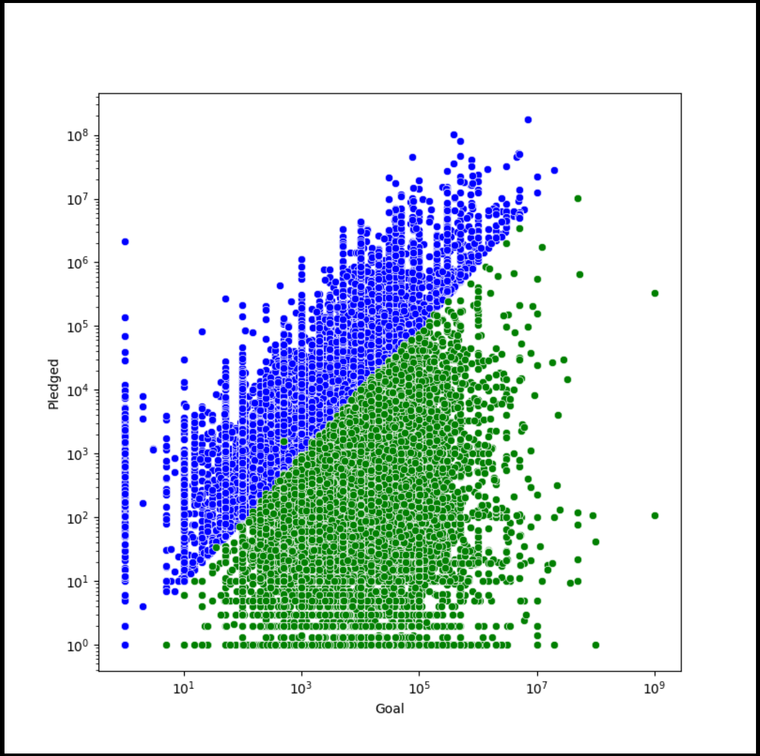
Exploratory Data Analysis



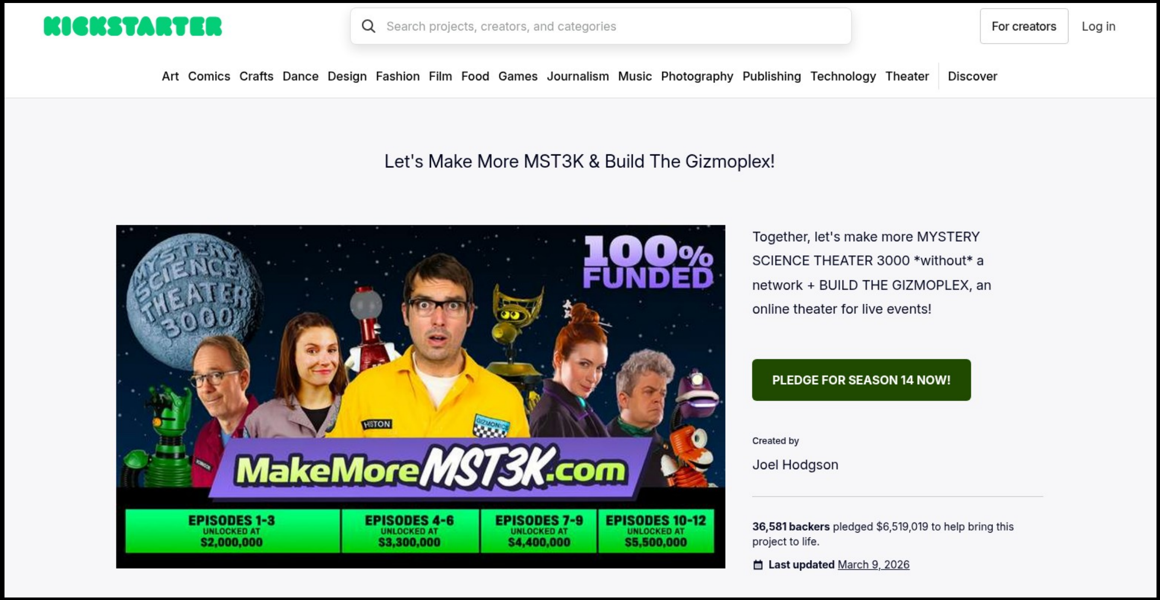
Word cloud based on common terms from Kickstarter project descriptions

Exploratory Data Analysis

- Goal vs Outcome
  - Successful campaigns in blue
  - Failed campaigns in green
- Note the some projects that had a goal of $1 and received millions, as well as vice versa



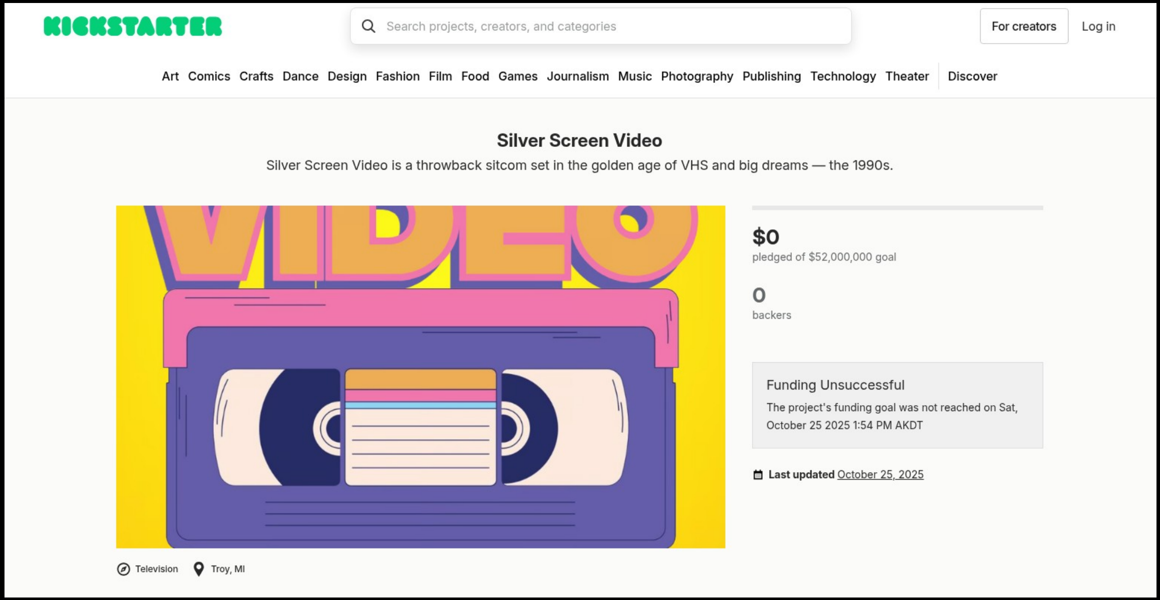
Exploratory Data Analysis



A notable project

- Mystery Science Theater 3000
- Goal: $2M USD, Pledge: $6.5M USD

Exploratory Data Analysis

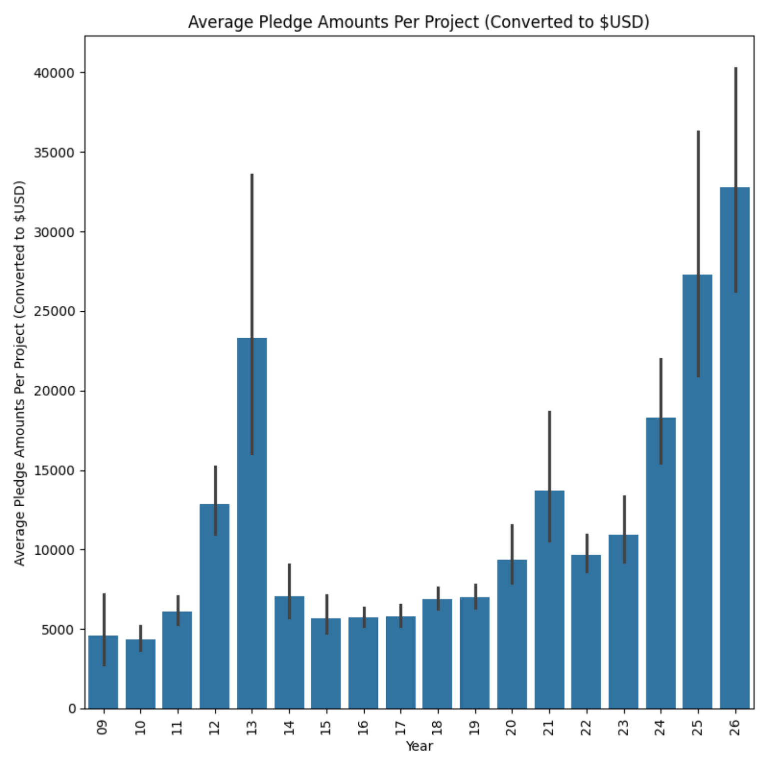


A notable project

- Silver Screen Video
- Goal: $52M USD, Pledge: $0 USD

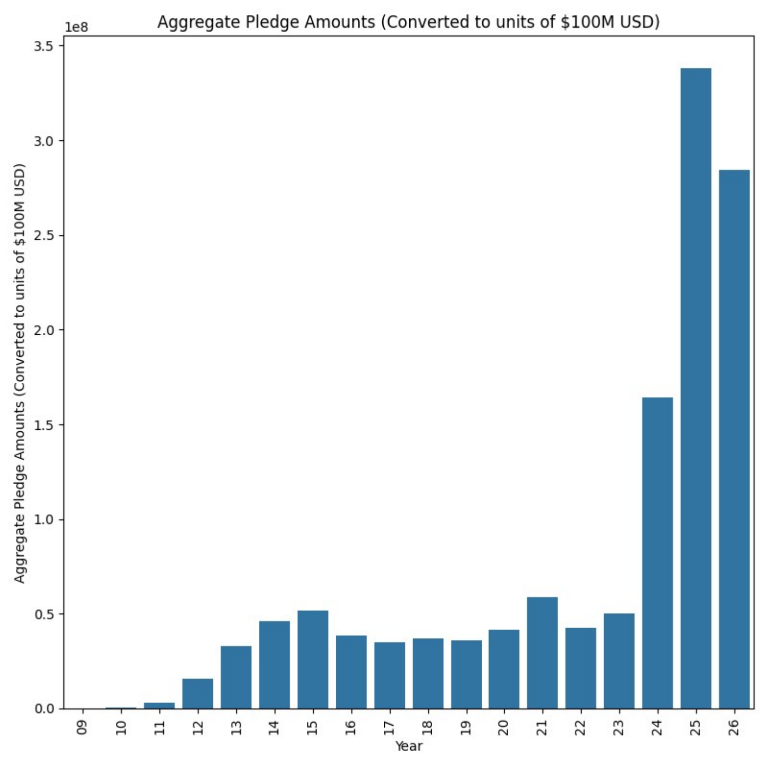
Exploratory Data Analysis

- Average pledged amounts per project are increasing
- (Note that this analysis could be subject to selection bias)



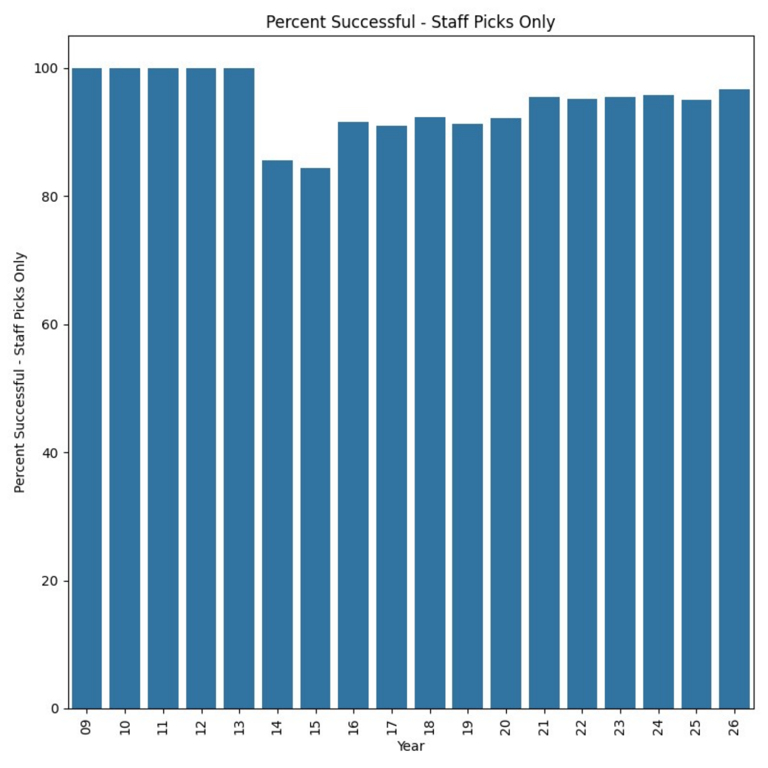
Exploratory Data Analysis

- Total pledged amounts are also increasing
- (Note possible selection bias)



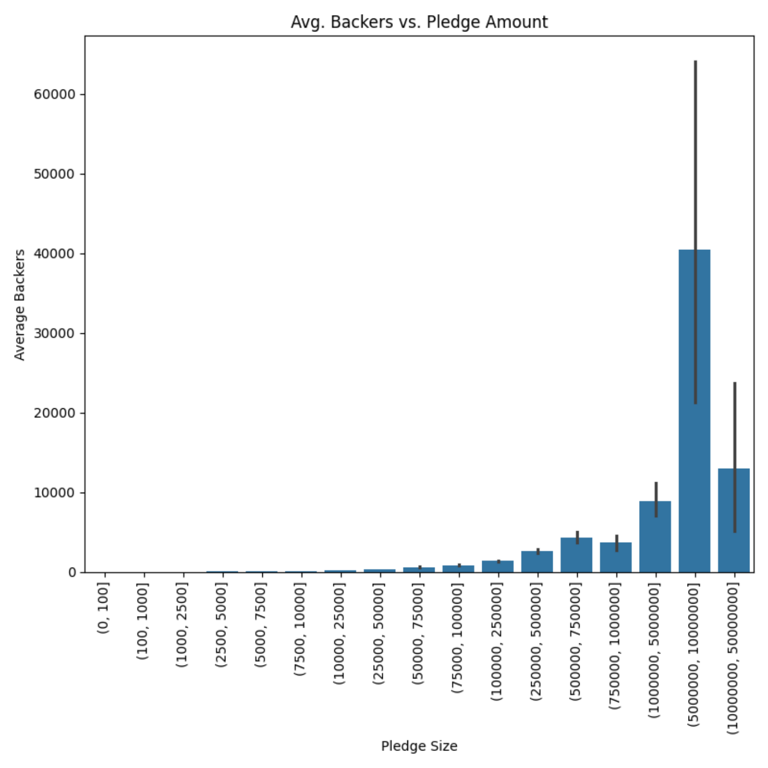
Exploratory Data Analysis

- Kickstarter projects that earn a “Staff Pick” highlight have an 80% chance or more of success



Exploratory Data Analysis

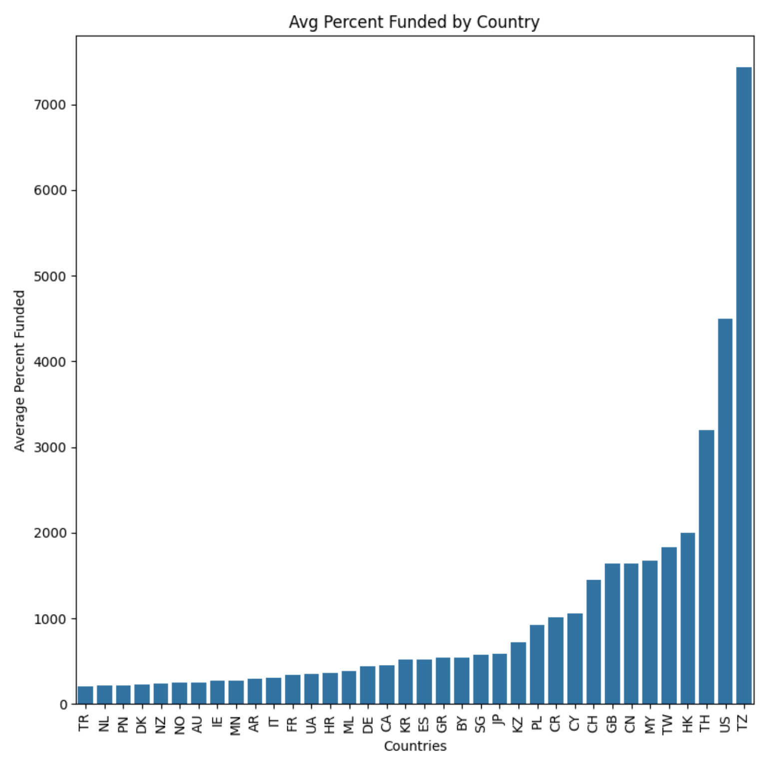
- Note that a project can earn as much as $100K without requiring more than 1000 backers
- Even 350 backers can raise $25K
- Seven-figure sums, however, require tens of thousands of backers



Exploratory Data Analysis

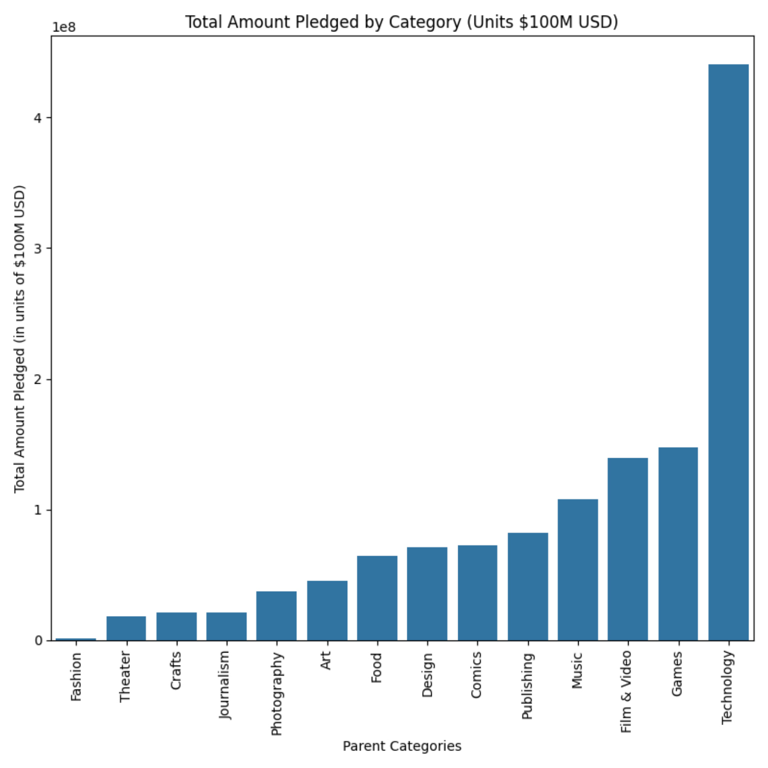
- Among successful projects, Tanzania is surprisingly the highest overachieving country, with an average of over 7000% project funding

This is likely due to a limited number of campaigns and skewed data



Exploratory Data Analysis

- The most lucrative category by far is Technology, followed by Games, Film & Video, and Music

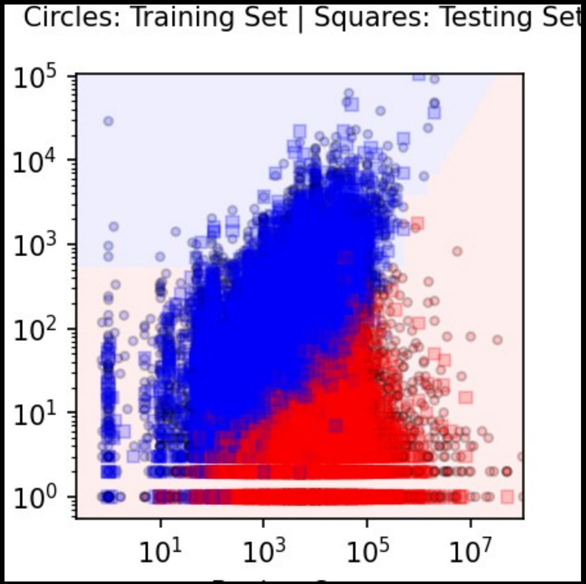


Exploratory Data Analysis

- Actionable Insights
  - Research how many committed donors beforehand
    - If only a few, keep goal at $100 or below
    - Considerable sums of $25000 < require 350+ backers

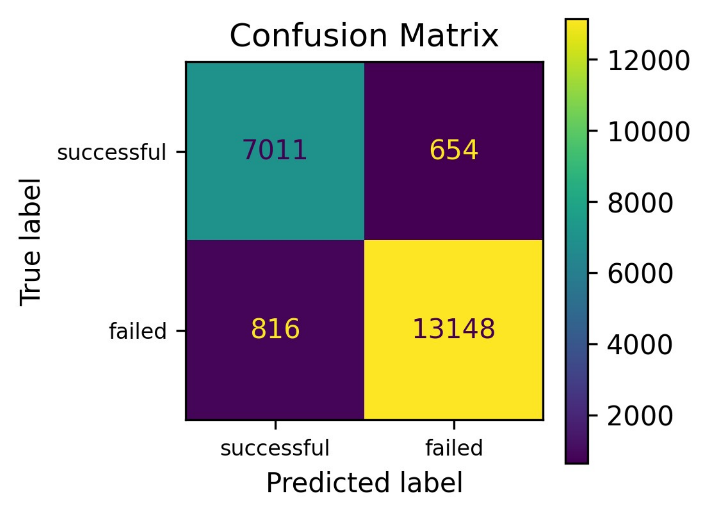
Predictive Modeling

- Logistic Regression Classification Decision Boundary Graph:
  - Successful campaigns in red
  - Failed campaigns in blue
  - x-axis: goal
  - Y-axis: pledge
- Note that successful campaigns are usually less ambitious



Predictive Modeling

- Logistic Regression Confusion Matrix
  - Overall accuracy: 0.92
- Logistic Regression model can likely provide a reasonably accurate prediction
- Notable features include:
  - Goal amount, expected backers, currency, location, category



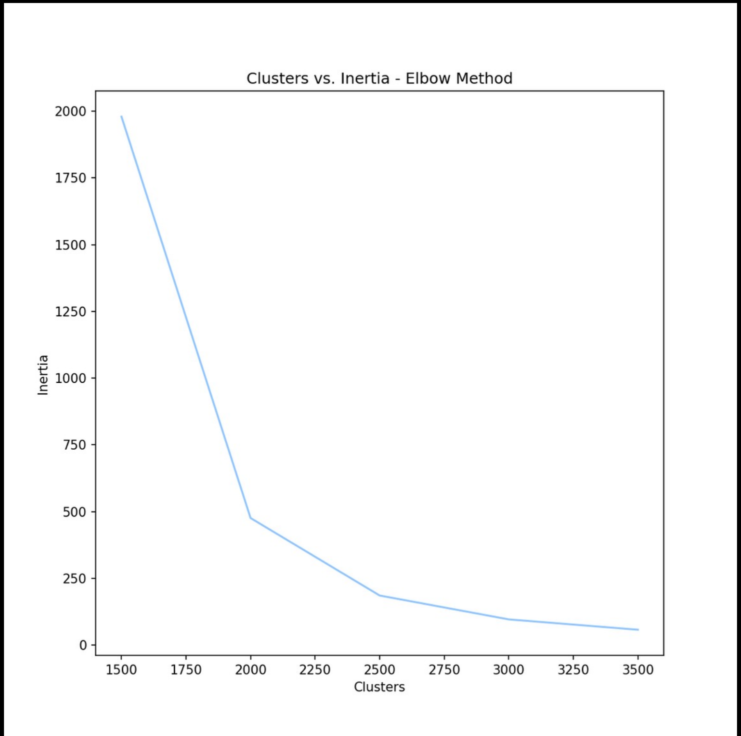
Predictive Modeling

K-Means Clustering

Total data rows: 86K

Inertia Elbow method:

Ideal number of clusters is likely 2250



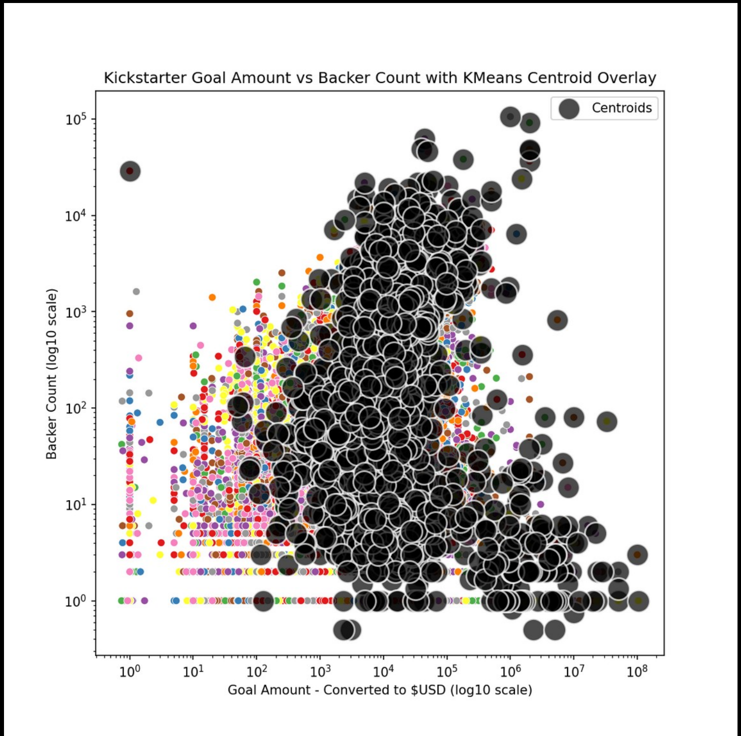
Predictive Modeling

Successful K-Means Cluster

Use case:

A Kickstarter candidate can put their parameters into the model and find similar projects in the same predicted cluster

This provides a useful tool for research and development



Conclusion

Recommendations and Insights:

- Keep project simple and low cost
- Research your number of committed supporters beforehand
- Limit your goal to a number that can be achieved
- Use K-Means Clustering to find similar projects and research their working models

To see a more complete version, visit:

- https://github.com/blue-santa/kickstarter-campaign-analysis
- https://bluesanta.io

Image Credits

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